

Coding & Solution

Date	03 October 2026
Team ID	17
Project Name	Fitbuddy-AI Fitness plan Generator
Department	Computer Technology
Institution / University	Madras Institute Of Technology / Anna University
Academic Year / Semester / Section	3 / 5 / Batch A
Faculty Mentor / Guide	Jayachitra V P
Team Leader	Bharath Jeyakkumar S
Sprint 1 Start / End	28-09-2026 / 28-09-2026
Sprint 2 Start / End	29-09-2026 / 30-09-2026
Sprint 3 Start / End	01-10-2026 / 02-10-2026
Team Roles	Abdul Wahith M — Backend & AI Integration; Kamalesh T — UI / Frontend Development; Bilu Besto H — Validation, Feedback & Reliability; Bharath Jeyakkumar S — Team Leader / Testing & Quality Assurance; Prasannavelan K — Database, Documentation & Demo Support
Maximum Marks	4 Marks

Milestone 1: Backend Foundation

- `app/main.py` creates the FastAPI application, mounts CSS/static content and initializes the database.
- `app/routes.py` contains the HTML routes and JSON API.
- `app/schemas.py` validates username, user ID, age, weight, goal, intensity and feedback.
- `app/database.py` persists users and plans using SQLAlchemy/SQLite.

Milestone 2: AI Integration

- `gemini_client.py` provides shared Gemini client behavior.
- `gemini_generator.py` generates seven-day workout plans.
- `gemini_flash_generator.py` generates nutrition/recovery tips.
- `updated_plan.py` revises a plan from natural-language feedback.
- `mock_ai.py` supplies deterministic responses for testing/demo mode.

Milestone 3: User Interface

The supplied application provides Jinja2 pages for input, results, feedback and the all-users/admin view.

Milestone 4: API and Persistence

The API includes /api/generate-workout, /api/submit-feedback, /api/nutrition-tip, /api/users, /api/users/{user_id} and /health.

Milestone 5: Testing and Reliability

The project includes pytest tests, isolated test database setup, mock AI, AI error handling, model fallback logic and a static fallback for nutrition tips.

Team implementation areas

Member	Main development contribution
Abdul Wahith M	FastAPI architecture, Gemini client integration and API flow
Kamalesh T	Templates, CSS and user-facing result/feedback/admin views
Bilu Besto H	Pydantic validation, feedback revision and failure handling
Bharath Jeyakkumar S	Automated tests and test verification
Prasannavelan K	SQLAlchemy persistence, user/plan management and setup/demo support